

Quasi-experimental research and spillover effects on Ethereum Merge

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Abstract

On September 15th, 2022, Ethereum transitioned from Proof of Work to Proof of Stake blockchain. The time-series Difference-in-Differences model is employed by taking the Bitcoin network as the control group while Ethereum as the treatment group to identify the effects of the protocol change. Vector AutoRegression models are used to identify the dynamic relationship between Ethereum and Bitcoin. Despite Ether's price dropping 10% initially, the daily returns and total fees in USD are not significantly different after Ethereum Merge, while trading volumes and various network factors in the Ethereum platform indicate significant differences compared to Bitcoin.

Keywords Ethereum Merge; Cryptocurrency; Difference in Differences (DiD); Spillover effects;

Vector AutoRegression (VAR); Proof of Stake (PoS)

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1. Introduction

Bitcoin, Litecoin, and many other cryptocurrencies generate the consensus of the block, employing the economic protocol known as Proof-of-Work (PoW). This system involves time and energy-intensive

operations (Cong et al., 2021). de Vries (2018) states that Bitcoin, the largest cryptocurrency, consumes electricity as much as in Austria and Ireland as a country annually. Participants and investors in the Ethereum network discuss environmental concerns and decide to implement the system change from PoW to Proof of Stake (PoS) blockchain. On December 1st, 2020, Ethereum launched a separate blockchain partaking PoS. After this date, two Ethereum platforms coexist. On September 15th, 2022, the original Ethereum platform with PoW merged into a PoS-based platform. Although the price of Ether dropped 10% on the day of completion, the change was meant to bring more stability and value to the network.

In this study, we analyze the effect of the Ethereum Merge on September 15th, 2022, employing two quasi-experimental methods--Difference in Differences (DiD) and Vector AutoRegression (VAR), and comparing two cryptocurrencies (Ethereum and Bitcoin). Figure 1 shows the standardized price movements for Bitcoin and Ethereum, showing both cryptocurrencies move in a similar direction, although Bitcoin prices tend to fluctuate more than Ether. Figure 2 shows the recent movements from August to November 2022, demonstrating Ether's price drop right after the Merge, and there is straightforward evidence of the spillover effect on Bitcoin. The correlation matrices of daily returns between Bitcoin and Ethereum are shown in Table 1 and Table 2, indicating the high connectedness. However, the association gets weaker before Ethereum Merge and gets stronger afterward. The rest of the paper is organized as follows: Section 2 presents the data, Section 3 introduces the methodology, Section 4 discusses the results, and Section 5 states the conclusion.

Place Tables 1 and 2, Figures 1 and 2 here

2. Data

We collect the price and volume data from Coinbase and CoinMarketCap. We conduct experiments, discuss the results using the Coinbase data, and note any differences from CoinMarketCap data.

To examine the effects of Ethereum Merge on September 15th, 2022, we collect the data two months before and after the event (July 15th to November 15th, 2022). The descriptive statistics for daily returns from Bitcoin and Ethereum are shown in Table 3, showing that the average returns are higher after Ethereum

Merge for both Bitcoin and Ether. The skewness also changes in a positive direction. In addition to the daily returns and volume changes, we also analyze various network factors. Cryptocurrency returns are more closely connected to the cryptocurrency network factors than production factors (Liu & Tsyvinski, 2021). The list and description of selected network variables are illustrated in Table 4. We collect the network factor data from CoinMetrics. Figure 3 and Figure 4 show the data for each variable for Bitcoin and Ethereum before and after Ethereum Merge.

Place Tables 3 and 4, Figures 3 and 4 here

3. Methodology

We employ two quasi-experimental techniques to examine the effect of Ethereum Merge, DiD, and VAR. We employ the DiD model to evaluate the counterfactual of the Ethereum Merge on the Ethereum network. The DiD model requires stable unit treatment value (exchangeability and positivity). Without Ethereum Merge, the difference between Ethereum and Bitcoin is constant over time. The composition of intervention and comparison groups is stable for repeated cross-sectional design. The initial hypothesis is that there are no spillover effects for the Bitcoin network since the Merge happens only in the Ethereum network. The DiD model without delayed effect is the following:

$$Y = \beta_0 + \beta_1 * [\textit{Time}] + \beta_2 * [\textit{Intervention}] + \beta_3 * [\textit{Time} * \textit{Intervention}] + \beta_4 * [\textit{Covariates}] + \varepsilon \quad (1)$$

The other method to examine the effect of Ethereum Merge is the VAR model. The VAR model considers a bi-directional relationship, examines the short-term dynamics between Bitcoin and Ethereum, and treats both variables symmetrically as endogenous. In the VAR model, determining the lag length is the critical input. We examine four criteria (AIC, HQ, SC, and FPE) and prioritize the lag section based on BIC measurement. BIC tends to choose a smaller number of lags than others, and BIC is preferred because the smaller number indicates the fitness of the lag and is suitable for VAR models (Clark & McCracken, 2007; Hyndman & Athanasopoulos, 2018). We show the result of lag length selection for each criterion in Table 5.

Place Table 5 here

The VAR estimation model has two types: VAR in levels and differences. "VAR in levels" is used to forecast the data with stationarity. If the data does not show stationarity, then we take the differences of the data to fit a VAR model (VAR in differences). Figure 3 and Figure 4 show the comparison of Bitcoin and Ethereum before and after the Ethereum Merge. Bitcoin variables show a more cyclical movement (non-stationary) nature, while the Ethereum variables tend to be steadier over time (stationary). Therefore, we use both estimation models and analyze the outputs. The estimation models for VAR in levels in equation 2 and in difference in equation 3 for forecasting and prediction are as follows:

$$Y_t = A_1 Y_{t-1} + \dots + A_p Y_{t-p} + CD_t + U_t Y_t \quad (2)$$

$$\Delta Z_t = B_1 \Delta Z_{t-1} + \dots + B_p \Delta Z_{t-p} + CD_t + U_t \Delta Z_t \quad (3)$$

where Y_t is a $K \times 1$ vector of endogenous variables, and U_t assigns a spherical disturbance term of the same dimension. ΔZ_t is the difference between endogenous variables. The coefficient matrices $A_1, \dots, A_p$ and $B_1, \dots, B_p$ are of dimension $K \times K$. In addition, either a constant or a trend can be included as deterministic regressors, centered seasonal dummy variables, and exogenous variables. The term CD_t , if there is no seasonal pattern, the seasonal dummy variable is equal to null by setting the type of argument to the corresponding value and setting season to the desired frequency (integer), and providing a matrix object for exogen, respectively.

We analyze spillover effects from Ethereum Merge by evaluating a set of variables--cryptocurrency returns, volumes, and network factors, such as the number of transactions, active supply, active and new address count, and total fees.

4. Results

Ether's price dropped by 10% on the day of the Merge. However, DiD analysis indicates non-significance in daily returns compared to the Bitcoin market. All the DiD results are shown in Table 6. The choice of the encoding scheme for intervention and periods determines the coefficients of a linear regression model. The sign of the variables can also affect the sign of coefficients in the results. Thus, the coefficients in the table are absolute values. If we choose -1 as the value of the independent variable for control and 1

for treatment, then we have a negative coefficient and a positive coefficient. The values for "before" and "after" periods also affect the magnitude of coefficients of independent variables for periods. There are two groups of intervention and two groups of periods, thus four values of interaction terms should be used for the experiment. We choose the values of encoding intervention and period based on this basic principle. Figure 3 and Figure 4 show the time-series data for each variable. Considering the means of each cryptocurrency and the changes caused by the Ethereum Merge, the signs of coefficients can be estimated. The trading volume shows that Ethereum significantly increased the average trading volume compared to Bitcoin after Ethereum Merge. Regarding the network factors, the variables--transaction, transfer, and active address counts show a similar pattern that the difference between Ethereum and Bitcoin variables becomes larger after Ethereum Merge. The variables, such as new address count, new funded address, and total fee in USD, show a difference between Ethereum and Bitcoin variables before, but Ethereum Merge does not cause the structural change between them. PoS protocol eliminates all the mining activities in the Ethereum platform, and the dynamics of active supply between Bitcoin and Ethereum have changed. Thus, the difference in active supply between Bitcoin and Ethereum becomes more prominent after the Merge. On the other hand, the addresses that hold the top 1% values in the network show a significantly increased value compared to Bitcoin after the Merge.

Place Table 6 here

For VAR analyses, we consider both the VAR in levels and in differences models analyzing the dynamic relationship and causality between Ethereum and Bitcoin after the Ethereum Merge. The outputs from VAR in differences (Table 7) show more robust and generate better results than VAR in levels (Table 8) in terms of the size of coefficients. Overall, lag length tends to be shorter, and the degree of cointegration appears to be lower after Ethereum Merge. For daily returns and trading volumes, both Bitcoin and Ethereum show evidence of stronger cointegration after the Ethereum Merge. The cointegration between Ethereum and Bitcoin has decreased for the transaction and transfer counts. For transaction count, the lag

length is also reduced to one day. For active and new address count, the spillover effect for the Bitcoin network is evident that cointegration has increased vastly after Ethereum Merge.

Place Tables 7 and 8 here

For the newly funded addresses, the lag length has been reduced to one day, and the size of cointegration has also dropped significantly, especially for the Ethereum platform after the Merge. Ethereum's active supply is strongly cointegrated with the Bitcoin network before the Merge, but now it shows higher cointegration to its network. Ethereum's total fee shows the cointegration to the Bitcoin network before the Merge, but the cointegration is much higher afterward. For the address balance of more than \$1 variable, the cointegration gets higher for Bitcoin and Ethereum. The value in the top 1% addresses shows an exception that the lag length becomes a lot longer after Ethereum Merge, from one to four days. Cointegration becomes more substantial, and both cryptocurrencies cointegrate stronger with Bitcoin's lagged values.

We also use the data from CoinMarketCap to analyze the effects of Ethereum Merge and examine the dynamic relationship between Bitcoin and Ethereum (Table 9 & Table 10). Although the coefficients are slightly different, the main results still hold. The trading volumes from Coinbase are a fraction of the total volumes shown from CoinMarketCap, and the data from CoinMarketCap shows that Ethereum Merge has affected the difference in trading volumes between Bitcoin and Ethereum.

Place Tables 9 and 10 here

5. Conclusion

We analyzed the effects of the Ethereum Merge on September 15th, 2022, using two quasi-experimental methods (DiD and VAR). Daily returns did not significantly change the relationship between Bitcoin and Ethereum. However, trading volume and various network factors, including the number of transactions, transfers, and active addresses, indicate the change in the dynamic relationship between Bitcoin and Ethereum. Results also indicate the spillover effects for the Bitcoin network. The total fee paid

in USD does not show that the new PoS protocol has successfully decreased the fee in Ethereum compared to Bitcoin. From the outputs of VAR in differences, the daily returns of Bitcoin and Ethereum show higher cointegration on the Ethereum network after Ethereum Merge. For some variables, such as total fee in USD, address balance over \$1, and the values in the top 1% addresses show higher cointegration from the Bitcoin network. Although the structure of daily returns has not changed, evidence suggests that the dynamics of how the existing and new crypto investors invest in Ethereum and Bitcoin after Ethereum Merge have changed.

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Figure 1 Standardized Price from 2016 to 2022

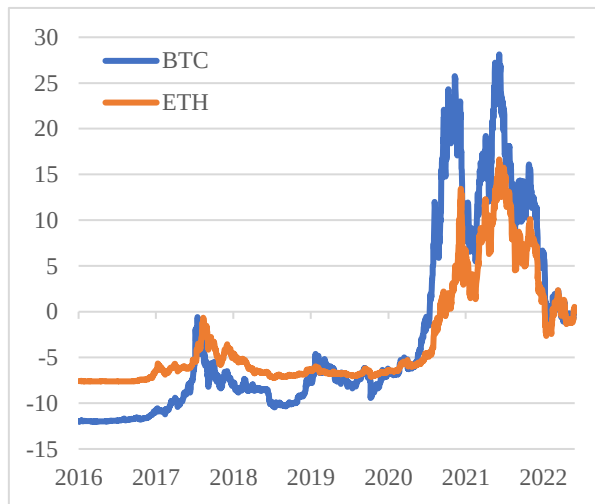


Figure 2 Standardized Price from August to October 2022

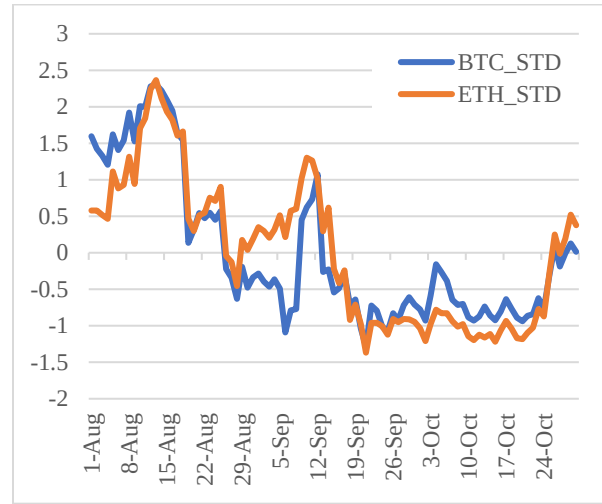


Table 1 Daily return correlation

	Corr.
2016-2022	93.4%
2016-2021	92.6%
2022	97.3%

Table 2

		Bitcoin	
		Before	After
Ethereum	Before	82.6%	
	After	93.4%	

before and after the Merge

Table 3 Descriptive statistics for daily returns before and after Ethereum Merge

Stats	Bitcoin		Ethereum	
	Before	After	Before	After
Mean	0.0091	-0.2393	0.5972	-0.3096
1st Qrt	-1.0446	-1.3466	-1.8834	-1.7811
Median	-0.2817	-0.5662	0.0282	-0.5493
3rd Qrt	1.3434	1.1459	2.6705	1.6969
IQR	2.3880	2.4925	4.5539	3.4780
SD	3.3966	3.2125	5.1180	4.9769
Skewness	-0.0649	-1.0678	0.4752	-0.2761
Kurtosis	3.1136	7.7262	2.5310	4.9654
t-stat	0.0209	-0.5865	0.9113	-0.4898

Table 4 Variable and description

Variable	Description
Transaction Count	The sum count of transactions in that day
Transfer Count	The sum count of transfers that interval
Total Fees (USD)	The sum USD value of all fees paid to miners, transaction validators, stakers and/or block producers at that interval.
Active Supply	The sum of unique native units held by accounts that transacted at least once up to that interval. Native units that transacted more than once are only counted once.
Address Balance >\$1	The sum of all native units being held in addresses whose balance was \$1 or greater at the end of that day.
Active Address Count	The sum count of unique addresses that were active in the network (either as a destination or source of a ledger change) that day
New Address Count	The sum count of all unique addresses that were newly created at that interval.
New Funded	The sum count of all unique addresses that were newly created and funded (e.g. have a non-zero balance) that interval
Addresses count	
Value in the Top 1% of Addresses	The sum of all native units held by the richest 1% addresses at the end of that time interval.

Table 5 Lag selection for the VAR model

Variable	Period	Model	BIC	AIC	HQ	FPE	Variable	Period	Method	BIC	AIC	HQ	FPE
Returns	Before	Level	1	1	1	1	New funded addresses	Before	Level	2	4	2	4
		Diff	1	1	1	1			Diff	2	4	2	4
	After	Level	1	1	1	1		After	Level	1	1	1	1
		Diff	1	1	1	1			Diff	1	1	1	1
Trading volume	Before	Level	1	4	4	4	Active supply	Before	Level	1	1	1	1
		Diff	1	4	4	4			Diff	1	1	1	1
	After	Level	1	2	1	2		After	Level	2	3	2	3
		Diff	1	2	1	2			Diff	2	3	2	3
Transaction count	Before	Level	2	4	4	4	Total fee (USD)	Before	Level	1	2	1	2
		Diff	2	4	4	4			Diff	1	2	1	2
	After	Level	1	2	1	2		After	Level	1	2	1	2
		Diff	1	2	1	2			Diff	1	2	1	2
Transfer Count	Before	Level	1	4	2	4	Address balance >\$1	Before	Level	1	1	1	1
		Diff	1	4	2	4			Diff	1	1	1	1
	After	Level	1	2	2	2		After	Level	1	2	2	2
		Diff	1	2	2	2			Diff	1	2	2	2
Active address count	Before	Level	1	2	2	2	Value in top 1% addresses	Before	Level	1	2	1	2
		Diff	1	2	2	2			Diff	1	2	1	2
	After	Level	1	1	1	1		After	Level	4	4	4	4
		Diff	1	1	1	1			Diff	4	4	4	4
New address count	Before	Level	1	4	4	4							
		Diff	1	4	4	4							
	After	Level	1	1	1	1							
		Diff	1	1	1	1							

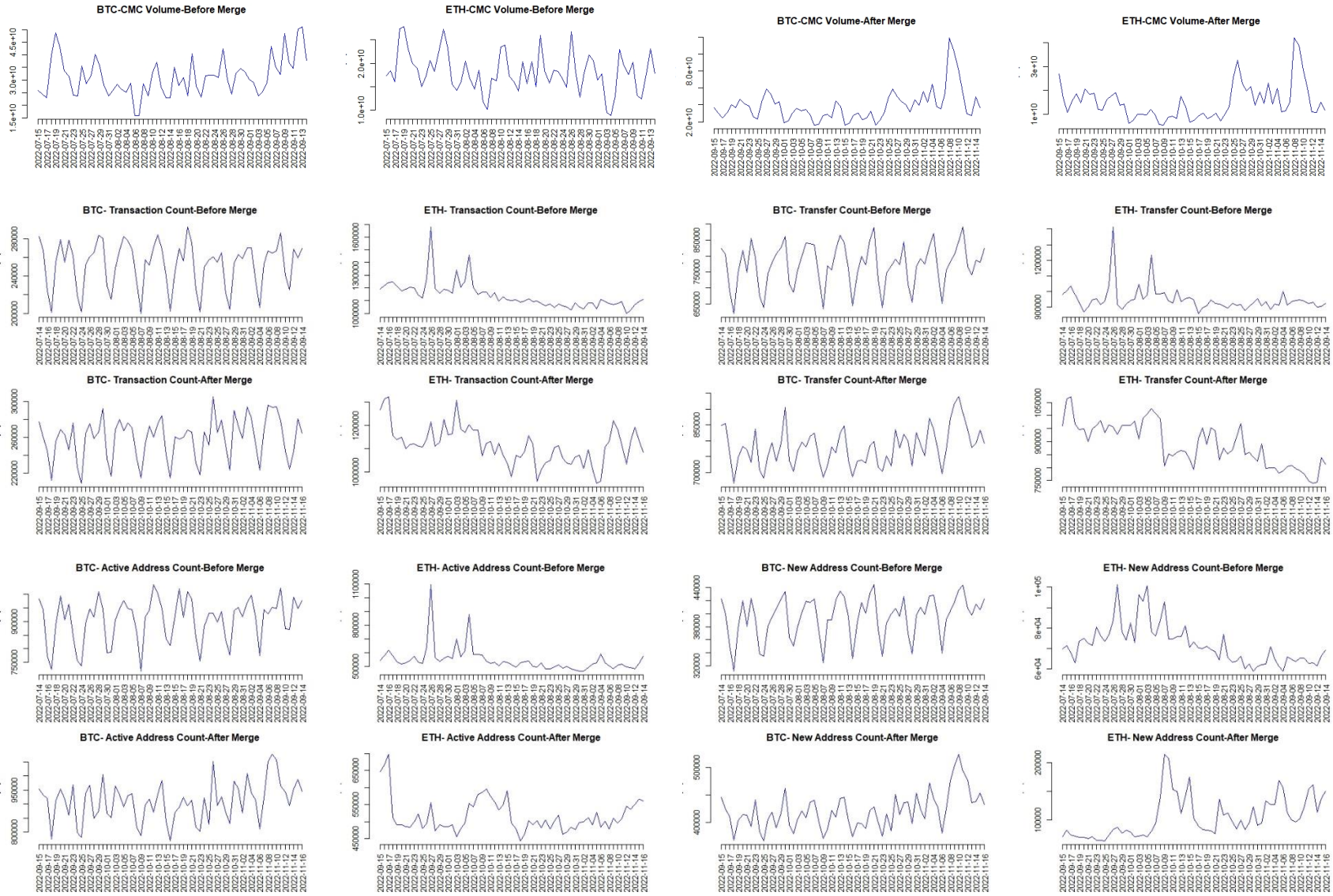


Figure 3 Comparison between Bitcoin and Ethereum before and after Ethereum Merge (1/2)

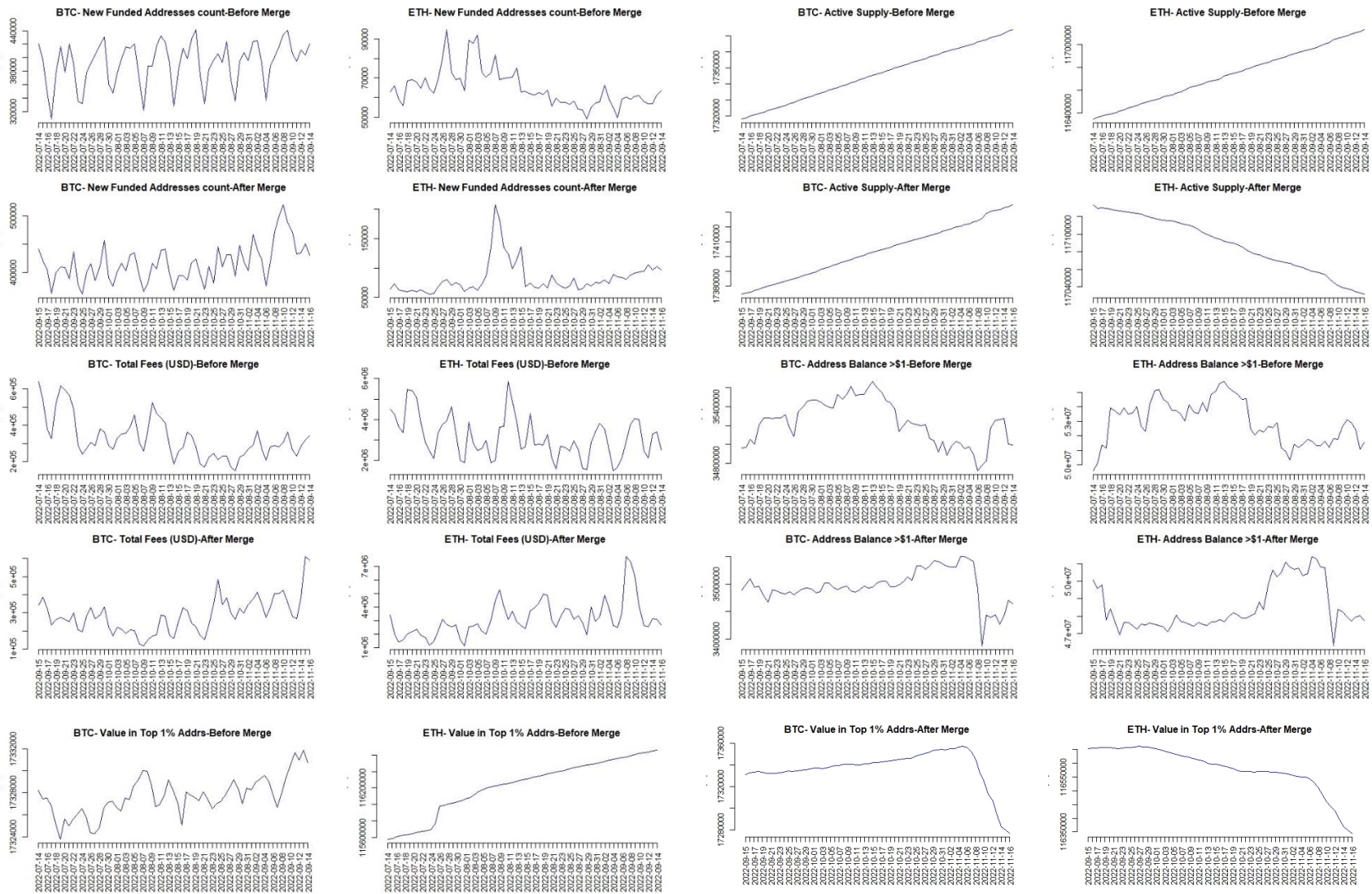


Figure 4 Comparison between Bitcoin and Ethereum before and after Ethereum Merge (2/2)

Table 6 DiD Outputs

	Model 1	Model 2		
	Diff in Treat	Diff in Treat	Diff in Periods	Interaction†
Returns	0.270	0.306	0.609	0.341
Trading volumes in \$1M	19.41	40.2	51.79	61.45*
Transaction count	872636***	444456***	9378	16276*
Transfer count	143469***	88520***	25801***	33571***
Active address count	375722***	179218***	14569	17286*
New address count	319622***	161518***	29235***	3413
New funded addresses	330361***	163606***	21188***	3148
Active supply	99534997***	49692457***	207626***	150083***
Total Fees (USD)	2828312***	1404823***	22565	18667
Address balance over \$1	15620872***	8855389***	2361889***	2089906***
Value in the top 1% addresses	99067962***	49445796***	185898***	176370***

This table reports the absolute value of the coefficient of each variable. *, **, and *** denote significance levels at the 10%, 5%, and 1% levels based on the standard t-statistics.

†Interaction term: Difference in treatment and control * Difference in periods

Table 7 VAR Model in Differences

Variable	Period	DV	BTC Lag 1	ETH Lag 1	BTC Lag 2	ETH Lag 2	Variable	DV	BTC Lag 1	ETH Lag 1	BTC Lag 2	ETH Lag 2	
Returns	Before	BTC	-0.73	0.03			New funded addresses	Before	BTC	-0.19	1.24	-0.62	1.18
		ETH	-0.10	-0.72				After	ETH	0.04	-0.46	-0.02	-0.33
	After	BTC	-0.39	-0.16					BTC	-0.25	-0.38		
		ETH	0.40	-0.92					ETH	0.07	0.08		
Change in volume	Before	BTC	-0.98	0.21			Active supply	Before	BTC	0.16	0.00		
		ETH	-0.08	-0.65					ETH	4.10	0.01		
	After	BTC	-0.42	0.07				After	BTC	0.18	-0.10		
		ETH	0.43	-0.90					ETH	-0.28	0.41		
Transaction count	Before	BTC	-0.07	0.05	-0.62	0.04	Total fee (USD)	Before	BTC	0.05	0.02		
		ETH	0.44	-0.56	-0.71	-0.59			ETH	0.72	-0.14		
	After	BTC	-0.33	0.09					BTC	0.02	0.00		
		ETH	-0.17	-0.18					ETH	-2.63	0.15		
Transfer count	Before	BTC	-0.01	0.11			Address balance >\$1	Before	BTC	0.08	-0.01		
		ETH	0.07	-0.61					ETH	-0.13	-0.03		
	After	BTC	-0.22	0.08				After	BTC	-0.42	0.12		
		ETH	0.20	-0.30					ETH	-0.88	0.10		
Active address count	Before	BTC	-0.05	0.02			Value in top 1%	Before	BTC	0.05	0.00		
		ETH	0.11	-0.49					ETH	-0.07	0.25		
	After	BTC	-0.42	-0.69				Addresses	After	BTC	0.37	0.35	0.91
		ETH	0.10	-0.27					ETH	0.26	1.13	1.41	-1.23
New address count	Before	BTC	-0.10	0.94					BTC				
		ETH	0.03	-0.61				DV	Lag 3	Lag 3	Lag 4	Lag 4	
	After	BTC	-0.34	-0.44				BTC	-0.62	0.37	-0.09	-0.03	
		ETH	0.13	0.05				ETH	-1.89	1.17	0.34	-0.22	

Table 8 VAR model in Levels

			BTC	ETH	BTC	ETH				BTC	ETH	BTC	ETH
Variable	Period	DV	Lag 1	Lag 1	Lag 2	Lag 2	Variable	DV	Lag 1	Lag 1	Lag 2	Lag 2	
Returns	Before	BTC	-0.09	0.06			New funded addresses	Before	BTC	0.44	0.40	-0.49	-0.08
		ETH	-0.09	-0.04		ETH			-0.03	0.53	-0.06	0.09	
Change in volume	After	BTC	0.13	-0.12			After	BTC	0.53	-0.01			
		ETH	-0.07	-0.03		ETH		0.10	0.79				
	Before	BTC	-0.49	0.29		Active supply	Before	BTC	0.93	0.00			
		ETH	-0.09	-0.11				ETH	1.51	0.90			
After	BTC	0.00	0.16		After	BTC	1.09	-0.16	-0.28	0.09			
	ETH	0.25	-0.19			ETH	-0.12	1.55	0.32	-0.67			
Transaction count	Before	BTC	0.34	0.02	-0.66	-0.02	Total fee (USD)	Before	BTC	0.85	0.85		
		ETH	-0.10	0.38	-1.11	-0.02			ETH	0.88	0.71		
Transfer count	After	BTC	0.28	0.05			After	BTC	0.97	0.01			
		ETH	-0.67	0.67		ETH		-2.19	1.01				
	Before	BTC	0.42	0.07		Address balance >\$1	Before	BTC	0.82	0.02			
		ETH	0.04	0.22				ETH	0.51	0.76			
Active address count	After	BTC	0.43	-0.11			After	BTC	0.51	0.12			
		ETH	0.06	0.64		ETH		-1.16	1.12				
	Before	BTC	0.50	-0.06		Value in top 1%	Before	BTC	0.67	0.00			
		ETH	0.06	0.31				ETH	-0.42	0.98			
New address count	After	BTC	0.23	-0.37		Addresses	After	BTC	1.28	0.28	0.71	-0.77	
		ETH	0.08	0.66				ETH	0.15	1.97	1.52	-2.05	
	Before	BTC	0.44	0.23				BTC		ETH	BTC	ETH	
		ETH	-0.02	0.39			DV	Lag 3	Lag 3	Lag 4	Lag 4		
After	BTC	0.52	0.02			BTC	-1.29	0.73	0.20	-0.25			
	ETH	0.26	0.76			ETH	-2.91	1.94	1.12	-0.86			

Table 9 DiD outputs for Daily Returns using CMC data

	1st model	2nd stage		
	Diff in Treat	Diff in Treat	Diff in Periods	Interaction
Returns	0.2696	0.3027	-0.6041	0.3358
Trading Volume in \$1M	18410***	6570***	2243	5299***

Table 10 VAR outputs for Daily Returns using CMC data

			Bitcoin Lag 1	Ethereum Lag 1
VAR in Difference	Before	BTC	-0.70	0.01
		ETH	-0.08	-0.72
	After	BTC	-0.43	-0.14
		ETH	0.34	-0.87
VAR in Levels	Before	BTC	-0.07	0.05
		ETH	-0.08	-0.03
	After	BTC	0.09	-0.10
		ETH	-0.12	0.01